mutual funds have even been launched by HF firms, seeking to branch out to mom-and-pop investors. (This money also tends to be more sticky than institutional HF investors.) New exchange-traded funds are specializing in some of these factor risk premiums. Many of the risk exposures that HFs bring to the table can be had more cheaply than 2/20, and many of these strategies are attracting more (and lower fee) copycats.[50] More capital piling in brings its own risks, too, as more investors may be chasing the same strategies. That could lead to more events like the August 2007 quant meltdown. As Andrew Lo says, "The whole hedge-fund industry is a series of crowded trades."[51]

There is ample room for innovation. At the time of writing in 2013, there are no large-scale, off-the-shelf factor portfolios for volatility, liquidity, or momentum that are global, diversified across asset classes, designed with low enough turnover to be very cheap (20–30 basis points), and available to ordinary investors. S&P 500, Russell 2000, and similar index funds cost 5 to 10 basis points for ordinary investors and nothing (literally, because of stock lending) for institutions, but the fees on factor portfolios will be higher because they dynamically trade. Some large investors (with tens to hundreds of billions of dollars) have created their own factor portfolios, but small asset owners can't. It is encouraging to see some asset management companies, including HF firms, pushing in this direction. Some HF firms have even introduced factor funds, but at prices much more expensive than 20 to 30 basis points. (However, the fees are generally much lower than the standard 2/20.) If factors can be commoditized, these HFs are at risk of being locked in a downward spiral of lower fees—good news for asset owners.

It's taking a while, though. Why don't we see dirt cheap factor funds now?

The First Factor Fund

We take low-cost, equity index funds for granted, but it wasn't obvious they were going to succeed when they started in the 1970s.[52] Traditional active managers, especially mutual fund families, didn't want to introduce index funds for fear of cannibalizing their higher-fee active funds, especially since active mutual funds underperform the market (see chapter 16). Offering index funds must have felt like throwing in the towel—like admitting you have no skill, and what manager wants to think that? Oldrich Vasicek, one of the early developers of the index fund said, "They thought we were crazy. . . . 'You just want to buy whatever garbage happens to be traded?' "[53]

[50] This includes "hedge fund replication" strategies (see Kat and Palaro 2005), but also pure factor portfolios in the spirit of chapter 14.

[51] Strasburg, J., and S. Pulliam, "Pack Mentality Grips Hedge Funds," *Wall Street Journal*, Jan. 14, 2011.

[52] The history of the index fund is outlined in Bernstein (1992) and MacKenzie (2006).

[53] Cited by MacKenzie (2006). Vasicek also wrote one of the most highly cited papers in fixed income, putting forward the Vasicek (1977) term structure model.